

Chapter 120 — Aging and Geriatric Urology — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Age alone ≠ surgical risk**; frailty, function, comorbidity decide
- **Fried frail** = ≥3 of 5; 1–2 = pre-frail
- **Fried weight loss**: >10 lb or 5% in 1 year
- **Bathing lost first, feeding last** (activities of daily living, ADLs)
- **Asymptomatic bacteriuria (ASB)** → **no antibiotics**; except pregnancy, before urological instrumentation
- **USPSTF (US Preventive Services Task Force) grade D**: ASB screening; PSA (prostate-specific antigen) screening in men >70
- **Life expectancy <10 yr** → **non-curative** prostate plan
- **Radical cystectomy at ≥75**: outcomes comparable to younger patients
- **Beers**: **avoid antimuscarinics** in dementia, delirium
- **Vaginal oestrogen safe**; systemic HRT (hormone replacement therapy) started ≥60 → harm > benefit

OPENING; DEMOGRAPHICS OF AGING

- **Geriatric** = ≥65, but chronological age a poor descriptor
- **UN (United Nations) alternative**: define by years of life remaining
- **Nascher, 1909**: first used "geriatrics" clinically
- **UN Decade of Healthy Ageing**: 2021–2030, WHO (World Health Organization)-led
- **Only ophthalmology, cardiology** exceed urology in geriatric volume
- **≥65**: fastest-growing; ~doubles to 94.7 M by 2060
- **2034**: over-65s outnumber children (US first)
- **≥85 triples**: 6 → 19 M by 2060

BIOLOGY AND PRINCIPLES OF AGING

- **Key driver: cellular senescence** (replication stops)
- **Hayflick limit (1961)**: telomere-driven replicative arrest
- **Senescence triggers**: telomeres; also ROS (reactive oxygen species), radiation, DNA damage, unfolded protein response
- **SASP** (senescence-associated secretory phenotype): inflammation, tumorigenesis; spreads senescence
- **WHO healthy ageing** = functional ability enabling wellbeing
- **Functional ability** = intrinsic capacity × environment; 5 abilities

CLINICAL EVALUATION

- **First step**: ability to function independently
- **Homeostenosis**: age-related narrowing of functional reserve
- **Chronic disease**: ~4 in 5 over-65s ≥1; ~1 in 4 ≥4
- **BADLs (basic ADLs)**: bathe, dress, groom, toilet, feed, ambulate
- **IADLs (instrumental ADLs)**: cook, clean, transport, laundry, finances, phone, medicines
- **Screen**: "Can you shower or bathe independently?"
- **Gradient**: independent > lose IADL > lose BADL → ↓ life expectancy, ↑ cost
- **Self-rated poor health** → predicts ADL/IADL decline over 3 yr
- **Occupational therapy** → restores independence, mobility
- **Worse surgical outcome**: long operation, poor ASA (American Society of Anesthesiologists) score, urgent/emergent case
- **Chronological age**: rarely an independent surgical predictor
- **"Clearing" for surgery** = **misnomer**; optimise instead
- **Box 120.1 (American College of Surgeons/American Geriatrics Society, 2012)**: 13-item checklist — only cardiac, pulmonary are organ items
- **Cardiac work-up**: ACC/AHA (American College of Cardiology/American Heart Association) non-cardiac surgery algorithm
- **SPAQI (Society for Perioperative Assessment and Quality Improvement, 2018)**: frailty screen before elective major surgery → diagnostic assessment → CGA

- **CGA** (comprehensive geriatric assessment): geriatrician-led; **no standard definition**
- **CGA sequence**: frailty → performance (360° turn, 3 chair rises, pen, jacket) → albumin, creatinine, BNP (brain natriuretic peptide) → FEV₁ (forced expiratory volume in 1 s)

CLINICAL EVALUATION — GERIATRIC SYNDROMES

- **List**: frailty, depression, delirium, falls, incontinence, polypharmacy, vision, hearing (+ pressure ulcers)
- **Model**: synergistic, multifactorial — not linear single-cause
- **Lifelines Cohort**: ≥1 syndrome → ↑ new chronic disease, independent of confounders
- **Dependent care** ↑: balance, cognitive impairment; **not** chronic pain
- **Frailty prevalence**: 7–10% at 65 → 25% >85
- **Fried five**: weight loss, weak grip, slow gait, low activity, exhaustion
- **Deficit accumulation index (DAI)**: not predictive postoperatively
- **Comorbidity vs disability vs frailty**: co-existing disease vs cannot self-care vs ↓ reserve

CLINICAL EVALUATION — BEERS (TABLE 120.1)

- **NSAIDs** (non-steroidal anti-inflammatory drugs): GI (gastrointestinal) risk age >75; PPI (proton pump inhibitor)/misoprostol if >7 days
- **COX-2 inhibitors**: avoid in HF (heart failure), CrCl (creatinine clearance) <30
- **Tramadol**: SIADH (inappropriate antidiuretic hormone secretion); CrCl <30 → avoid extended-release
- **Edoxaban**: avoid CrCl <15 or >95; reduce dose at 15–50
- **Dabigatran option** only if <75 yr
- **Ciprofloxacin**: avoid with theophylline; ↑ INR (international normalised ratio) with warfarin; reduce dose at CrCl <30
- **Macrolide exempt**: azithromycin
- **Nitrofurantoin**: avoid CrCl <30, chronic use; OK in moderate impairment
- **TMP-SMX** (trimethoprim-sulfamethoxazole): ↓ dose at CrCl 15–29; avoid <15; check K⁺ at 5 days
- **Tricyclics**: avoid in men with BPH (benign prostatic hyperplasia); doxepin listed >6 mg/day
- **Benzodiazepines**: avoid with ≥2 other CNS (central nervous system) drugs, opioids
- **Testosterone**: only confirmed symptomatic hypogonadism
- **Desmopressin** → hyponatraemia; treat nocturia cause; 5-ARI (5α-reductase inhibitor) for BPH
- **Vaginal oestradiol** <25 mcg twice weekly after breast cancer (with discussion)

- **Vasomotor:** SSRI (selective serotonin reuptake inhibitor; not paroxetine), SNRI (serotonin–noradrenaline reuptake inhibitor), gabapentin
- **Rating codes:** A = avoid; C = caution; H = CMS (Centers for Medicare & Medicaid Services) high-risk

CLINICAL EVALUATION — GSM, FI, PREVENTION

- **GSM** (genitourinary syndrome of menopause): coined 2014; 27–84% of postmenopausal individuals
- **GSM → urgency, frequency, UI (urinary incontinence);** vaginal oestrogen may relieve UI (urgency UI)
- **Oestrogen deprivation (breast cancer), ADT (androgen deprivation), 5-ARI → ↑ OAB** (overactive bladder)
- **Hormone-sensitive cancer:** not absolute contraindication to vaginal oestrogen → consult oncologist
- **Caruncle:** first-line topical oestrogen; excision rarely needed
- **Postmenopausal bleeding →** endometrial cancer in 9%
- **FI** (faecal incontinence): most dreaded syndrome; <30% of women report it → ask
- **FI prevalence:** 14.4% US; 17% community women >85; 43% nursing home
- **Antihypertensives ↓** mortality >80
- **Testosterone Trials:** no cognitive benefit; modest sexual benefit
- **ASPREE (aspirin):** no disability-free survival gain; ↑ mortality

CLINICAL EVALUATION — INCONTINENCE TREATMENT

- **Antimuscarinics:** 20–70% better than placebo; poor persistence
- **Persistence ↑:** M3-selective (solifenacin, darifenacin, trospium)
- **Trospium:** quaternary amine → less BBB (blood–brain barrier) penetration
- **β3-agonist:** dry mouth < placebo; constipation = placebo
- **Mirabegron:** metoprolol interaction; ↑ BP
- **Tolterodine high dose →** QT prolongation, torsades
- **After starting:** consider PVR (post-void residual); never exceed maximum dose
- **Nursing-home UI:** ~½ short-stay, ~¾ long-stay
- **New UI after admission =** unmasked, pre-existing
- **Catheter colonisation:** up to 98%

CLINICAL EVALUATION — TABLE 120.2

- **Fever:** >100°F or >2°F above baseline
- **Loeb, catheterised:** ≥1 of fever, new CVA (costovertebral angle) tenderness, rigors, acute delirium
- **Loeb, no catheter:** dysuria, or fever + ≥1 new urinary symptom
- **NHSN** (National Healthcare Safety Network): culture ≥10⁵ CFU (colony-forming units) + no other site + signs

- **NHSN leukocytosis:** >14,000/mm³; left shift >6% bands or ≥1500/mm³
- **NHSN repeated fever:** >99°F

CLINICAL EVALUATION — UTI AND BACTERIURIA

- **Lifetime UTI:** >50% women; 12% men
- **ASB =** positive culture, **any pyuria**, no symptoms
- **ASB prevalence:** up to 22% older women; ~10% older men
- **ASB: AGS Choosing Wisely —** one of original five
- **Transplant ASB:** no evidence of graft failure
- **First task:** symptomatic UTI vs OAB vs ASB

CLINICAL EVALUATION — CANCER

- **Prostate cancer** ≈11% of US cancer treatment cost
- **Bladder cancer:** greatest risk factor = age; 4th commonest, 8th cancer death (US men)
- **Restaging TURBT** (transurethral resection) for T1 intermediate/high risk → underused in elderly; improves survival
- **BCG** (bacille Calmette–Guérin) less effective in elderly (immunosenescence)

END OF LIFE CARE; ADDITIONAL RESOURCES; KEY POINTS

- **First step:** goals-of-care conversation; family may disagree
- **Palliative ≠ withdrawal;** aggressive symptom control
- **4 Ms:** Mobility, Medication, Mentation, what Matters
- **GAYF:** "Geriatrics at Your Fingertips" (AGS, annual)
- **POP** (pelvic organ prolapse): age alone no bar to surgery
- **Elder abuse:** urologists mandatory reporters in most places

CLASSIC TRAPS

- **Trap:** old → too risky for surgery — **Actually:** frailty, ASA, urgency predict; age rarely independent
- **Trap:** confused resident + positive culture = UTI — **Actually:** no causal link; ASB untreated
- **Trap:** pyuria makes it a UTI — **Actually:** ASB defined regardless of pyuria
- **Trap:** count deficits for frailty — **Actually:** DAI not predictive; use Fried
- **Trap:** aspirin protects healthy elderly — **Actually:** ASPREE ↑ mortality
- **Trap:** breast cancer bars vaginal oestrogen — **Actually:** not necessarily; consult oncology
- **Trap:** nitrofurantoin banned in any CKD — **Actually:** avoid at CrCl <30 only
- **Trap:** caruncle bleeding = benign — **Actually:** exclude endometrial, cervical cancer
- **Trap:** new nursing-home UI = new disease — **Actually:** often unmasked
- **Trap:** HRT is cardio- or neuroprotective — **Actually:** neither; harm if started ≥60